

Codebook

Technical documentation for the refreshed Generic/Specific binary model

52 Ranked questions	28 Selected questions	29.98 Selected minutes	2.561 Average Pi
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What changed in this refresh

- The final category model now uses only Generic and Specific.
- The previous Financial Crisis label is retained only as a historical traceability field.
- The 0.7 threshold is applied to a normalized Ri score so the split is mathematically meaningful.
- The final CSV is rebuilt from the raw integrated source during every production run.

Workflow Design

How the raw integrated CSV becomes the final scored client package

Pipeline

```
PSID_Ranked_Questions_Katrina_Integrated.csv -> normalize_ranked_questions() -> parse_keywords() / tag_keywords() /
extract_constructs() -> compute_utility() / compute_burden() / Ri -> assign_binary_categories() -> compute_augmented_scores() ->
select_for_time_budget() -> PSID_Ranked_Questions_Final.csv + summary + PDFs
```

Core files

File	Role
PSID_Ranked_Questions_Katrina_Integrated.csv	Raw integrated source bank
PSID_NLP_Crisis_Module_Structure.py	Reusable helpers and thresholds
generate_psid_artifacts.py	Production rebuild script
PSID_Ranked_Questions_Final.csv	Authoritative scored output
PSID_NLP_Crisis_Module_Final.ipynb	Validation notebook

Functions and Formulas

Simple descriptions of the functions that matter for review

Key functions

Function	What it does
build_ranked_dataset()	Rebuilds the final dataframe from the raw source
assign_binary_categories()	Computes threshold score and Generic/Specific label
compute_augmented_scores()	Adds Pi, redundancy, construct bonus, and portability
select_for_time_budget()	Greedy selection under the 30-minute cap
suggest_deployable_wording()	Converts historical items into deployable wording
build_variable_names()	Creates stable GEN_ and SPC_ identifiers

Core formulas

- $U_i = \text{sum}(\text{keyword weights})$
- $B_i = \max(0.10 * \text{word_count} + 0.20 * \text{complexity}, 0.1)$
- $R_i = U_i / B_i$
- $ri_threshold_score = (R_i - \min(R_i)) / (\max(R_i) - \min(R_i))$
- Generic if $ri_threshold_score \geq 0.70$ else Specific
- $P_i = \text{augmented_utility} / [B_i * (1 + 0.65 * \text{redundancy_scaled})]$

Worked Example

Current top Generic item under the refreshed binary model

Calculation table

Field	Value	Calculation
Ui	3.400	$0.80 + 0.80 + 0.90 + 0.90$
Bi	0.600	$\max(0.10 * 6 + 0.20 * 0.0, 0.1)$
Ri	5.667	$3.40 / 0.60$
ri_threshold_score	1.000	$(5.667 - 0.818) / (5.667 - 0.818)$
Category	Generic	$1.000 \geq 0.70$
Pi	7.860	Enhanced priority after IDF, construct, richness, and redundancy adjustments

Top Generic rows

Variable	Recommended wording	Ri	Pi
GEN_LOSE_EARNINGS_PANDEMIC	Did you lose earnings because of the pandemic?	5.667	7.860
GEN_EXPERIENCED_FINANCIAL_DIFFIC	Have you experienced any financial difficulties because of the crisis?	5.667	7.187
GEN_RECEIVE_STIMULUS_PAYMENT_O	Did you receive a stimulus payment or other emergency government support?	4.667	6.328

Threshold and Interaction Notes

Why the binary split is normalized and how demographics behave

Raw-threshold issue

- Minimum raw Ri in the refreshed corpus is 0.818 and maximum raw Ri is 5.667.
- A direct raw Ri >= 0.7 rule would classify 52 of 52 rows as Generic.
- The production pipeline therefore applies the 0.7 rule to min-max normalized Ri.

Interaction summary

Group	Rows	Average Ri	Average Pi
Demographics only	3	1.354	2.189
Non-demographic crisis items	49	2.253	2.584

Interpretation: demographic items remain useful for traceability, but they carry less direct crisis signal than housing, trauma, aid, employment, or economic-loss items. That is why the deployable questionnaire excludes demographics even though the master questionnaire retains them.

Outputs and Benchmarks

Current client package outputs and refreshed production metrics

Deliverables

Deliverable	Purpose
Master_Questionnaire.xlsx	Internal review spreadsheet with all 52 rows
Deployable_Questionnaire.pdf	Selected deployable questionnaire excluding demographics
Codebook.pdf	Technical codebook
Final_Report.pdf	Stakeholder-facing report

Benchmark summary

Metric	Value
Rows	52
Selected rows	28
Selected minutes	29.98
Average Ri	2.201
Average Pi	2.561
Generic rows	3
Specific rows	49

